

Article

Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy

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Abstract: Quantization of neural networks has been one of the most popular techniques to compress models for embedded (IoT) hardware platforms with highly constrained latency, storage, memory-bandwidth, and energy specifications. Limiting the number of bits per weight and activation has been the main focus in the literature. To avoid major degradation of accuracy, common quantization methods introduce additional scale factors to adapt the quantized values to the diverse data ranges, present in full-precision (floating-point) neural networks. These scales are usually kept in high precision, requiring the target compute engine to support a few high-precision multiplications, which is not desirable due to the larger hardware cost. Little effort has yet been invested in trying to avoid high-precision multipliers altogether, especially in combination with 4 bit weights. This work proposes a new quantization scheme, based on power-of-two quantization scales, that works on-par compared to uniform per-channel quantization with full-precision 32 bit quantization scales when using only 4 bit weights. This is done through the addition of a low-precision lookup-table that translates stored 4 bit weights into nonuniformly distributed 8 bit weights for internal computation. All our quantized ImageNet CNNs achieved or even exceeded the Top-1 accuracy of their full-precision counterparts, with ResNet18 exceeding its full-precision model by 0.35%. Our MobileNetV2 model achieved state-of-the-art performance with only a slight drop in accuracy of 0.51%.

Keywords: quantization; neural networks; nonuniform; power-of-two scales; low-cost hardware



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1. Introduction

Quantization of neural networks dates back to the 1990s [1,2], where the discretization of models was a necessity to make their implementation feasible on the available hardware. More recently, neural networks became popular again because of the ImageNet challenge [3] and the availability of powerful GPU hardware. This breakthrough started a new area of research with hundreds of new potential applications. Today, neural networks are found in various electronic devices such as smartphones, wearables, robots, self-driving cars, smart sensors, and many others. The embedded electronic processors found in these applications are often very limited in capabilities to make them affordable, compact, and energy efficient in the case of battery-powered applications. Most neural networks are too large and require too many computations to be implemented directly on such processors, and therefore need to be compressed first. One of the most effective ways to reduce latency, storage cost, memory-bandwidth, energy efficiency, and silicon area among popular compression techniques such as model pruning [4] and network architecture search [5] is model quantization [6]. The quantization of neural networks is a frequently visited research topic with numerous publications that mostly focus on reducing the number of bits per weight or activation as much as possible in order to achieve high compression rates [7–11].

The commonly accepted method to achieve low-precision quantization introduces high-precision scale factors, and in some cases zero-points, to adapt the quantized values to the diverse weight and activation ranges present in full-precision neural networks. This